Machine Learning, Chapters 1–5: Symbols and Formulas

Symbol	What it is	Ch.
Data, inputs, outputs		
$x \in \mathcal{X} \subseteq \mathbb{R}^d$	one input: a vector of d *features*; x_j is feature j	2
$y \in \mathcal{Y}$	the output to predict: a number (regression) or a class label (classification)	1–2
X, Y	input and output seen as *random variables*: "a house drawn at random and its price"	2
$\mathbb{P}(X, Y)$, $\mathbb{P}(Y \mid X{=}x)$	joint distribution of the world; distribution of the output once the input is known	2
$\mathbb{E}[\cdot \mid X = x]$	average over that conditional distribution	2
$\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$	the training data: N examples, assumed i.i.d. draws from $\mathbb{P}(X, Y)$; $x_j^{(i)}$ = feature j of example i	2
$\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{val}}$, $\mathcal{D}_{\text{test}}$	the three splits: learn θ; choose hyperparameters; estimate final performance once	2–3
C; Δ^{C-1}	number of classes; the set of probability vectors over C classes (nonnegative, sum to 1)	2
Decisions and losses (Chapter 2)		
$a \in \mathcal{A}$	an *action*: a predicted number, a chosen class, "raise the alarm"	2
$\ell(y, a)$, ℓ_{dec}	a *loss*: cost of acting a when the truth is y (decision loss, set by the application)	2
ℓ_{train}	the training loss actually optimised (e.g. squared error, cross-entropy), often $\neq \ell_{\text{dec}}$	2
$f^*(x)$	the *Bayes rule*: best decision when $\mathbb{P}$ is known, $\arg\min_a \mathbb{E}[\ell(Y, a) \mid X{=}x]$	2
C_{FP}, C_{FN}	cost of a false positive / false negative; they set the decision threshold	2
k	number of neighbours in kNN (a hyperparameter)	2
Models, risk, training (Chapters 2–3)		
f_θ, $\theta \in \Theta$	a model with adjustable *parameters* θ (learned from data)	2
$\mathcal{H} = \{f_\theta : \theta \in \Theta\}$	the *hypothesis space*: every function the model can represent; choosing it is an inductive bias	2–3
λ	a *hyperparameter*: chosen by us on validation data, not learned (regularisation strength, k, depth, ...)	2–3
$R(\theta)$	*population risk*: expected loss over the whole world; what we want, unknowable	2
$\widehat{R}(\theta)$, $\widehat{R}_{\text{train}}$	*empirical risk*: average loss on the data we have; ERM minimises it	2
$\hat{f}_{\mathcal{D}}$	the model a fixed procedure produces when fed dataset $\mathcal{D}$ (a random object, since $\mathcal{D}$ is)	3
$G(\hat{f}_{\mathcal{D}})$	*generalization gap*: population risk minus training risk of the same fitted model	3
$\mathbb{E}_{\mathcal{D}}[\cdot]$, $\bar{f}(x)$	average over many possible training sets; the average prediction at x across them	3
$\text{Bias}(x)$, $\text{Var}(x)$	$\bar{f}(x) - f^*(x)$ (systematic error) and $\mathbb{E}_{\mathcal{D}}[(\hat{f}_{\mathcal{D}}(x) - \bar{f}(x))^2]$ (sample-to-sample spread)	3
ε, $\sigma^2(x)$	noise in Y around the best prediction, and its variance (irreducible)	3
$\Omega(\theta)$; $\|\theta\|_2^2$, $\|\theta\|_1$	a complexity *penalty*; the ridge (L_2, $\sum \theta_j^2$) and lasso (L_1, $\sum \lvert\theta_j\rvert$) penalties	3
$h \in \mathcal{C}$; $\hat{f}_h^{(-j)}$	one hyperparameter configuration from a candidate set; model with config h fitted on all folds except j	3
TP, FP, TN, FN	counts of true/false positives/negatives on a test set	3
Information theory (Chapter 1)		
$H(p)$	entropy of distribution p: $-\sum_c p_c \log p_c$; the irreducible floor of the cross-entropy	1
$H(p, q)$	cross-entropy of q relative to p: $-\sum_c p_c \log q_c$; the classification training loss	1–2
$\text{KL}(p \Vert q)$	Kullback–Leibler divergence $\sum_c p_c \log(p_c/q_c) \geq 0$, zero iff $q = p$ (Gibbs)	1
$H(Y \mid X)$	conditional entropy: the floor of the expected cross-entropy for a classifier	2
Classification (Chapter 4)		
$p(y{=}c \mid x)$; p_θ, $\hat{p}$	the *posterior* (true); a model's / a fitted model's estimate of it	4
$s_{\theta,c}(x)$	a real-valued *score* (logit) for class c; the score family fixes the boundary geometry	4
$\sigma(s)$	the *sigmoid* $1/(1 + e^{-s})$: one score $\to$ a probability in $(0, 1)$	4
$\text{softmax}(s)_c$	$e^{s_c} / \sum_{c'} e^{s_{c'}}$: C scores $\to$ C probabilities summing to 1; shift-invariant	4
$\pi_c = p(y{=}c)$	the class *prior*; MLE is the class frequency N_c/N	4
$p_\theta(x \mid y{=}c)$	the *class-conditional*: which inputs class c produces (generative route)	4
$\phi = (\pi, \theta)$	all parameters of a generative classifier: priors plus class-conditional parameters	4
$\mathcal{N}(x; \mu, \Sigma)$; $\hat{\mu}_c$, $\widehat{\Sigma}_c$	Gaussian density; fitted class mean and covariance (shared $\widehat{\Sigma}$ = LDA, separate = QDA)	4
N_c; n_{cj}; V	examples of class c; count of token j in class c's documents; vocabulary size	4
α	additive-smoothing pseudo-count (Laplace when $\alpha = 1$)	4
π_t, π_d	class priors at training time and at deployment time (prior shift)	4
Ensembles (Chapter 5)		
M; $\hat{f}_m(x)$, $\hat{p}_m(\cdot \mid x)$; $\hat{f}_{\text{ens}}$	number of models; prediction / posterior of model m; the ensemble prediction (average, vote or sum)	5
σ^2, ρ	error variance of one model; correlation between the errors of two models	5
$\mathcal{D}_m^*$; $\text{count}(i)$; $\mathcal{M}_i$	m-th bootstrap sample (N draws with replacement); times i was drawn; the out-of-bag models for i	5
$S_1, \ldots, S_J$; θ_j	the leaves (regions) of a tree and the constant predicted in each	5
$I(S)$; $\hat{p}_c(S)$	impurity of a node (Gini or squared error); fraction of class c in node S	5
m_{try}	candidate features drawn at random at each split of a random forest (= d gives bagging)	5
Δ_t; MDI_j; I_j	credit of split t; mean-decrease-in-impurity importance; permutation importance of feature j	5
$F_m(x)$; $T_m(x)$; η	boosted model after m rounds; the m-th tree; learning rate (shrinkage)	5
$r_m^{(i)}$	residual (squared loss) or pseudo-residual (negative gradient) of example i before round m	5
$w_{q,m}$; $S_{q,m}$; Q_m	output of leaf q of tree m; its region; number of leaves of tree m	5
$g^{(i)}$, $h^{(i)}$; G_q, H_q	first / second derivative of example i's loss w.r.t. its logit; their sums over leaf q	5
λ, γ (XGBoost)	penalty on squared leaf outputs; penalty per leaf (minimum gain to split)	5
z_m; $h_{\hat{\gamma}}$	stacking: base-model predictions; the fitted meta-model	5
$g_m(x)$, $a_m(x)$	mixture of experts: gate weight and gate score of expert m ($g = \text{softmax}(a)$)	5

When to use what.

- **Bias high** (train and val both bad): more capacity, better features, less regularisation.
- **Variance high** (train good, val bad): more data, regularise, fewer features, bag or random forest.
- **Little data, credible assumptions**: generative (LDA, Naive Bayes). **Much data**: discriminative, boosting.
- **Unequal costs**: keep the probabilities, move the threshold to $C_{\text{FP}}/(C_{\text{FP}} + C_{\text{FN}})$.
- **Unstable deep trees**: bagging / forest (parallel). **Structured residuals**: boosting (sequential, shallow, small η, early stopping).

Numbers worth knowing.

- Covariance parameters: $d(d+1)/2$ ($d = 100 \Rightarrow 5050$ per class).
- Bootstrap: $(1 - 1/N)^N \to 0.368$; $N = 10$: 0.349. Feature subsets: 2^d.

Formulas and rules

Each line: the formula, then what it says in words. All logs natural.

Decision theory (Ch. 2): the loss decides the best action

$f^*(x) = \arg\min_a \mathbb{E}[\ell(Y,a) \mid X{=}x]$ — Bayes rule: pick the action with the smallest expected loss at this x. Nothing is "best" until ℓ is chosen.

squared $\to \mathbb{E}[Y \mid x]$; absolute $\to$ median; asymmetric $\to$ quantile; zero–one $\to \arg\max_c p(y{=}c \mid x)$ — which summary of $Y \mid x$ each regression loss asks for; most probable class

declare 1 iff $p(x) > \dfrac{C_{\mathrm{FP}}}{C_{\mathrm{FP}} + C_{\mathrm{FN}}}$ — unequal costs move the threshold off $\frac{1}{2}$: costly misses $\Rightarrow$ lower threshold

$\mathbb{P}(\text{neighbour within } 0.1) = 1 - (1 - 10^{-d})^n$ — curse of dimensionality: local methods (kNN) starve as d grows; standardise features first

Risk, ERM, cross-entropy (Ch. 1–2)

$R(\theta) = \mathbb{E}[\ell(Y, f_\theta(X))]$; $\widehat{R}(\theta) = \frac{1}{N}\sum_i \ell(y^{(i)}, f_\theta(x^{(i)}))$ — population vs empirical risk; ERM: $\hat\theta = \arg\min_\theta \widehat{R}(\theta)$

$\mathbb{E}[\widehat{R}_{\text{train}}(\hat\theta)] \le R(\hat\theta)$ after fitting — training risk is honest for a *fixed* model, optimistic after the data chose it; nested families only lower it

$H(p,q) = H(p) + \mathrm{KL}(p\|q)$, $\mathrm{KL} \ge 0$ — cross-entropy is minimised exactly by $q = p$: it rewards calibrated probabilities

$\ell_{\mathrm{CE}} = -\log \hat p_{\theta,y}(x)$; $\mathbb{E}[\ell_{\mathrm{CE}}] = H(Y \mid X) + \mathbb{E}[\mathrm{KL}]$ — the training loss for classifiers; equals negative log-likelihood (maximum likelihood)

Generalization, bias–variance, regularisation (Ch. 3)

$\mathbb{E}_{\mathcal{D},Y|x}[(Y - \hat f_{\mathcal{D}}(x))^2] = \sigma^2(x) + \mathrm{Bias}(x)^2 + \mathrm{Var}(x)$ — noise (irreducible) + systematic error^2 + sample sensitivity; capacity trades bias for variance

$\min_\theta \widehat{R}(\theta) + \lambda\Omega(\theta)$ — regularised ERM $\equiv$ fitting inside a constrained (nested) family; λ chosen on validation

ridge 1-D: $\hat w = z/(1+\lambda)$; lasso 1-D: $\hat w = \mathrm{sign}(z)\max(|z| - \lambda, 0)$ — L_2 shrinks smoothly, keeps all features; L_1 soft-thresholds, sets coefficients exactly to 0

CV: $\frac{1}{K}\sum_j \widehat{R}_{\text{fold } j}(\hat f_h^{(-j)})$, then refit on all — every example validates once; selection (features, λ) must happen *inside* each fold or it leaks

$\text{Precision} = \frac{TP}{TP+FP}$; $\text{Recall} = \frac{TP}{TP+FN}$; $\text{Specificity} = \frac{TN}{TN+FP}$ — accuracy hides minority failure; balanced accuracy $= \frac{1}{2}(\text{Recall} + \text{Specificity})$

$F_1 = \frac{2PR}{P+R}$, $\min(P,R) \le F_1 \le \frac{P+R}{2}$ — harmonic mean: dragged toward the worse of the two

Discriminative classification (Ch. 4)

$\hat p(y{=}1 \mid x) = \sigma(s_\theta(x))$; $\sigma(-s) = 1 - \sigma(s)$; $\sigma(s) > \frac{1}{2} \iff s > 0$ — binary: one score; boundary is $s = 0$ (threshold t: $s > \log\frac{t}{1-t}$)

$\hat p(y{=}c \mid x) = \mathrm{softmax}(s)_c$; only $C-1$ contrasts matter — with $C = 2$ softmax is the sigmoid of the score difference

Generative classification (Ch. 4)

$p(y{=}c \mid x) = \dfrac{\pi_c\, p(x \mid y{=}c)}{\sum_{c'} \pi_{c'}\, p(x \mid y{=}c')}$ — Bayes' rule: compare prior-weighted likelihoods; the posterior is a softmax of $\log\pi_c + \log p(x \mid c)$

LDA: $s_c(x) = x^\top \widehat\Sigma^{-1}\hat\mu_c - \frac{1}{2}\hat\mu_c^\top\widehat\Sigma^{-1}\hat\mu_c + \log\hat\pi_c$ — MLE: $\hat\pi_c = N_c/N$, $\hat\mu_c =$ class mean, $\widehat\Sigma_c =$ class covariance (denominator N_c). Shared $\widehat\Sigma \Rightarrow$ linear boundaries; QDA (separate) $\Rightarrow$ quadratic; a covariance costs $d(d+1)/2$ parameters

Naive Bayes: $p(x \mid c) = \prod_j p(x_j \mid c)$ — conditional independence: score $= \log\pi_c + \sum_j \log p(x_j \mid c)$, one term per feature

$\hat p(\text{token } j \mid c) = \dfrac{n_{cj} + \alpha}{\sum_{j'} n_{cj'} + \alpha V}$; $\hat\sigma_j^2 \leftarrow \hat\sigma_j^2 + \lambda s_j^2$ — additive smoothing kills zero counts; variance smoothing kills zero variances

prior shift: generative swap $\hat\pi_t \to \pi_d$; discriminative odds $\times \frac{\pi_d/\pi_t}{(1-\pi_d)/(1-\pi_t)}$ — same class-conditionals, new priors; the discriminative fix needs calibrated probabilities

Ensembles (Ch. 5)

$\mathrm{Var}(\hat f_{\text{ens}}) = \sigma^2\left(\rho + \frac{1-\rho}{M}\right)$ — averaging M models: independent errors give σ^2/M; correlation leaves a floor $\rho\sigma^2$

Gini $I(S) = 1 - \sum_c \hat p_c(S)^2$; split credit $\Delta_t = N_t I(S_t) - N_L I(S_L) - N_R I(S_R)$ — trees: greedy impurity reduction; leaf predicts mean / median / majority

$\mathbb{P}(i \notin \mathcal{D}_m^*) = (1 - \frac{1}{N})^N \to e^{-1} \approx 0.368$ — bootstrap: 63.2% distinct in, 36.8% out; OOB error uses the $\approx 0.368M$ models that never saw i

random forest: bagging + m_{try} random candidates per split, start $\approx \sqrt{d}$ — feature randomisation lowers ρ faster than it raises σ^2; tune by OOB; add trees until OOB plateaus

$\text{MDI}_j =$ mean over trees of $\sum_{t \text{ on } j} \Delta_t/N$ (normalised); $I_j =$ OOB error after shuffling j − before — MDI: fast, biased to many-valued features; permutation: measures reliance; correlated features mask each other

$F_m = F_{m-1} + \eta T_m$, $F_0 = \bar y$; $r_m^{(i)} = -\partial\ell/\partial F(x^{(i)})$ — boosting: a sum of corrections fitted to the negative gradient; squared loss $\Rightarrow$ plain residuals

logit boosting: $g = \hat p - y$, $h = \hat p(1-\hat p)$; leaf $w = \dfrac{\sum(-g)}{\sum h + \lambda}$ — binary classification: boost the logit, Newton step per leaf

XGBoost: $w_q^* = -\dfrac{G_q}{H_q + \lambda}$, $\phi_q = -\frac{1}{2}\dfrac{G_q^2}{H_q + \lambda}$ — best leaf output and the loss it removes; $\mathcal{L}_m = \sum_q [G_q w_q + \frac{1}{2}(H_q + \lambda)w_q^2] + \gamma Q_m$

$\text{gain} = \frac{1}{2}\left[\dfrac{G_L^2}{H_L+\lambda} + \dfrac{G_R^2}{H_R+\lambda} - \dfrac{G^2}{H+\lambda}\right] - \gamma$ — split iff gain > 0; multi-class: K trees per round, $g_k = \hat p_k - y_k$, $h_k = \hat p_k(1-\hat p_k)$

stacking: $\hat f = h_\gamma(z_1, \dots, z_M)$ on out-of-fold z; MoE: $p(y \mid x) = \sum_m g_m(x)\, p_m(y \mid x)$ — stacking: one global learned combination (in-sample z leaks); MoE: input-dependent softmax gate, trained jointly with the experts